The Symmetric Three-Radical Pratt–Mason Inequality

An independent proof and a catalogue of applications

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Abstract

Let $f_n(x)$ be the multiplicative polynomial encoding associated with Pratt forests, let $h(n) = m_2(n) = \deg f_n$, and let $\rho(n) = \sum_{p|n} m_2(p) = \deg \operatorname{rad}(f_n)$. For a primitive additive triple $a + b = c$, three natural polynomial lifts of a, b, c are

$$G_a = f_c - f_b, \qquad G_b = f_c - f_a, \qquad G_c = f_a + f_b.$$

Writing $A_i = \deg \operatorname{rad}(G_i)$ and ordering $h(a), h(b), h(c)$ as $h_{(1)} \leq h_{(2)} \leq h_{(3)}$, we prove

$$\boxed{h_{(2)} + 2h_{(3)} \leq 2\rho(abc) + A_a + A_b + A_c - 3.} \tag{S3R}$$

The proof is a cyclic application of Mason–Stothers followed by an exact summation of three pairwise maxima. The rest of the note records consequences for radical growth, multiple roots, perfect powers, primitive factors, S-unit finiteness, Fermat–Catalan and Pillai-type equations, power sums, arbitrary Pratt coordinates, complete-front statistics, Hadamard–Pratt powers, ramification, and computation.

Contents

1 Framework and source status

For positive integers, the Pratt-coordinate framework supplies multiplicative polynomials $f_n \in \mathbb{Z}[x]$ with the following properties:

$$f_{mn} = f_m f_n, \qquad f_n(2) = n,$$
$$\deg f_n = m_2(n), \qquad \deg \operatorname{rad}(f_n) = \sum_{p|n} m_2(p).$$

The prime-index polynomials f_p are pairwise distinct irreducibles, so $\gcd(m,n) = 1$ implies $\gcd(f_m, f_n) = 1$. We use the abbreviations

$$h(n) := m_2(n) = \deg f_n, \qquad \rho(n) := \sum_{p|n} m_2(p) = \deg \operatorname{rad}(f_n).$$

Thus h is completely additive, whereas ρ depends only on the ordinary prime support.

Where the underlying proof first appears. Within the supplied manuscript collection, the one-sided inequality

$$\max\{h(u), h(v)\} \leq \rho(uv) + \deg \operatorname{rad}(f_u + f_v) - 1$$

first appears, with proof, as Theorem 18.1 on page 23 of *Pratt Coordinates for Natural Numbers and Monic Rational Polynomials*, Version 0.2, August 3, 2026 [1, Theorem 18.1, pp. 23–24]. That manuscript also identifies the addition defect as the obstruction to transferring the polynomial theorem directly to the classical integer *abc* problem [1, pp. 23–25]. The symmetric three-radical inequality *itself is not stated there*; it is derived and proved independently in Section 2 below by applying the same Mason argument to all three cyclic lifts.

2 The symmetric three-radical theorem

Let $a, b, c \in \mathbb{N}$ satisfy

$$a + b = c, \qquad \gcd(a,b) = 1, \qquad ab > 1.$$

The last condition excludes the exceptional all-constant polynomial relation arising from $(a,b,c) = (1,1,2)$. Since $a + b = c$ and $\gcd(a,b) = 1$, the three integers are pairwise coprime. Define

$$G_a := f_c - f_b, \qquad G_b := f_c - f_a, \qquad G_c := f_a + f_b,$$

and

$$A_a := \deg \operatorname{rad}(G_a), \qquad A_b := \deg \operatorname{rad}(G_b), \qquad A_c := \deg \operatorname{rad}(G_c).$$

The subscripts are chosen because

$$G_a(2) = a, \qquad G_b(2) = b, \qquad G_c(2) = c.$$

Put

$$R := \rho(a) + \rho(b) + \rho(c) = \rho(abc),$$

and order the three heights as

$$h_{(1)} \le h_{(2)} \le h_{(3)}.$$

Lemma 2.1 (Three cyclic Mason inequalities)**.** *Under the preceding hypotheses,*

$$\max\{h(b), h(c)\} \le \rho(b) + \rho(c) + A_a - 1, \tag{1}$$
$$\max\{h(a), h(c)\} \le \rho(a) + \rho(c) + A_b - 1, \tag{2}$$
$$\max\{h(a), h(b)\} \le \rho(a) + \rho(b) + A_c - 1. \tag{3}$$

Proof. We prove (1); the other two are cyclic variants. The identity

$$f_b + G_a = f_c$$

is a nonconstant polynomial relation. Because b and c are coprime, f_b and f_c are coprime. Moreover,

$$\gcd(f_b, G_a) = \gcd(f_b, f_c - f_b) = \gcd(f_b, f_c) = 1,$$

and similarly $\gcd(f_c, G_a) = 1$. Hence the three polynomials are pairwise coprime. Mason–Stothers gives

$$\max\{\deg f_b, \deg G_a, \deg f_c\} \le \deg \operatorname{rad}(f_b G_a f_c) - 1.$$

The three factors are pairwise coprime, so radical degrees add:

$$\deg \operatorname{rad}(f_b G_a f_c) = \deg \operatorname{rad}(f_b) + \deg \operatorname{rad}(f_c) + \deg \operatorname{rad}(G_a) = \rho(b) + \rho(c) + A_a.$$

Using $\deg f_n = h(n)$ and discarding the term $\deg G_a$ from the maximum yields (1). □

Theorem 2.2 (Symmetric three-radical Pratt–Mason inequality)**.** *Let $a+b = c$, $\gcd(a, b) = 1$, and $ab > 1$. Then*

$$\boxed{h_{(2)} + 2h_{(3)} \le 2\rho(abc) + A_a + A_b + A_c - 3.} \tag{S3R}$$

Proof. Add the three inequalities in Theorem 2.1. Every ρ-term occurs twice, so the sum of the right-hand sides is

$$2\big(\rho(a) + \rho(b) + \rho(c)\big) + A_a + A_b + A_c - 3 = 2\rho(abc) + A_a + A_b + A_c - 3.$$

For three ordered real numbers $h_{(1)} \le h_{(2)} \le h_{(3)}$, the sum of their three pairwise maxima is exactly

$$\max(h(a), h(b)) + \max(h(a), h(c)) + \max(h(b), h(c)) = h_{(2)} + 2h_{(3)}.$$

Substitution gives (S3R). □

Remark 2.3 (Why the coefficient pattern is exact)**.** The coefficient 2 on the largest height and the coefficient 1 on the second largest are not estimates: they are the exact multiplicities with which those heights occur among the three pairwise maxima. Thus no information is lost in the symmetrisation step.

3 Immediate radical and cancellation consequences

Throughout the remainder of the note, the hypotheses and notation of Theorem 2.2 are in force unless stated otherwise. Put

$$A_\Sigma := A_a + A_b + A_c.$$

Corollary 3.1 (Individual radical lower bounds)**.** *One has*

$$\begin{aligned} A_a &\geq \max\{h(b), h(c)\} - \rho(b) - \rho(c) + 1, \\ A_b &\geq \max\{h(a), h(c)\} - \rho(a) - \rho(c) + 1, \\ A_c &\geq \max\{h(a), h(b)\} - \rho(a) - \rho(b) + 1. \end{aligned}$$

Proof. Rearrange the three inequalities in Theorem 2.1. □

Corollary 3.2 (A dominant term forces two large radicals)**.** *Suppose $h(a) = H = h_{(3)}$. Then*

$$A_b \geq H - \rho(a) - \rho(c) + 1, \qquad A_c \geq H - \rho(a) - \rho(b) + 1,$$

and hence

$$A_b + A_c \geq 2H - 2\rho(a) - \rho(b) - \rho(c) + 2.$$

The analogous statements hold when b or c has maximal height.

Corollary 3.3 (Total radical growth)**.** *The total additive radical degree satisfies*

$$\boxed{A_\Sigma \geq h_{(2)} + 2h_{(3)} - 2\rho(abc) + 3.}$$

If every height is at least $\lambda h_{(3)}$ for some $0 \leq \lambda \leq 1$, then

$$A_\Sigma \geq (2 + \lambda)h_{(3)} - 2\rho(abc) + 3.$$

In particular, if $h(a) = h(b) = h(c) = H$, then

$$A_\Sigma \geq 3H - 2\rho(abc) + 3.$$

Proof. The first inequality is (S3R) rearranged. The second uses $h_{(2)} \geq \lambda h_{(3)}$. □

Corollary 3.4 (Bounded cancellation at infinity)**.** *For the three lift polynomials,*

$$\begin{aligned} \max\{h(b), h(c)\} - \deg(f_c - f_b) &\leq \rho(b) + \rho(c) - 1, \\ \max\{h(a), h(c)\} - \deg(f_c - f_a) &\leq \rho(a) + \rho(c) - 1, \\ \max\{h(a), h(b)\} - \deg(f_a + f_b) &\leq \rho(a) + \rho(b) - 1. \end{aligned}$$

Proof. For example, $\deg G_a \geq A_a$. Combine this with the first inequality of Theorem 3.1. The other cases are identical. □

4 Multiple roots, multiplicities, and perfect powers

For a nonzero polynomial $G \in \mathbb{Q}[x]$, define its multiple-root defect

$$\mu(G) := \deg G - \deg \operatorname{rad}(G) = \deg \gcd(G, G').$$

Set $D_i := \mu(G_i)$.

Theorem 4.1 (Triple multiple-root defect bound)**.** *One has*

$$\begin{aligned} D_a &\le \rho(b) + \rho(c) - 1, \\ D_b &\le \rho(a) + \rho(c) - 1, \\ D_c &\le \rho(a) + \rho(b) - 1, \end{aligned}$$

and consequently

$$\boxed{D_a + D_b + D_c \le 2\rho(abc) - 3.}$$

Proof. The full Mason inequality used in the proof of Theorem 2.1 gives

$$\deg G_a \le \rho(b) + \rho(c) + A_a - 1.$$

Subtracting A_a yields $D_a \le \rho(b) + \rho(c) - 1$. Sum cyclically. □

Corollary 4.2 (Multiplicity of an irreducible factor)**.** *If an irreducible $Q \in \mathbb{Q}[x]$ satisfies $Q^e \mid G_a$, then*

$$(e-1)\deg Q \le \rho(b) + \rho(c) - 1,$$

so

$$e \le 1 + \left\lfloor \frac{\rho(b) + \rho(c) - 1}{\deg Q} \right\rfloor.$$

In particular, every factor of degree at least $\rho(b) + \rho(c)$ occurs simply in G_a. The cyclic analogues hold for G_b and G_c.

Proof. The contribution of Q^e to $\mu(G_a)$ is $(e-1)\deg Q$. Apply Theorem 4.1. □

Corollary 4.3 (Perfect-power obstruction)**.** *Assume G_a is nonconstant and $G_a = H^k$ for some $k \ge 2$. Then*

$$\boxed{\max\{h(b), h(c)\} \le \frac{k}{k-1}(\rho(b) + \rho(c) - 1).}$$

The same conclusion holds if every irreducible factor of G_a occurs with multiplicity at least k.

Proof. Let $M = \max\{h(b), h(c)\}$. Since $G_a = H^k$,

$$A_a = \deg \operatorname{rad}(H) \le \deg H = \frac{\deg G_a}{k} \le \frac{M}{k}.$$

By Theorem 2.1, $M \le \rho(b) + \rho(c) + A_a - 1$. Therefore

$$M \le \rho(b) + \rho(c) - 1 + \frac{M}{k},$$

which rearranges to the stated bound. The k-full case uses $A_a \le \deg G_a/k$ in the same way. □

5 Factor counts, large factors, and primitive factors

Let $\omega_{\mathbb{Q}[x]}(G)$ denote the number of distinct nonconstant irreducible factors of G over $\mathbb{Q}$.

Corollary 5.1 (Many small factors or one large factor)**.** *Put*

$$L_a := \max\{h(b), h(c)\} - \rho(b) - \rho(c) + 1.$$

If all irreducible factors of G_a have degree at most d, then

$$\omega_{\mathbb{Q}[x]}(G_a) \geq \left\lceil \frac{\max\{0, L_a\}}{d} \right\rceil.$$

Conversely, if G_a has at most N distinct irreducible factors and $L_a > 0$, then it has an irreducible factor Q with

$$\deg Q \geq \frac{L_a}{N}.$$

Proof. By Theorem 3.1, $A_a \geq L_a$. If all factor degrees are at most d, then $A_a \leq d\,\omega_{\mathbb{Q}[x]}(G_a)$. If there are at most N factors, one of them has degree at least A_a/N. □

Corollary 5.2 (A Zsigmondy-type escape criterion)**.** *Let $\mathcal{T}$ be a finite set of irreducible polynomials and put*

$$D(\mathcal{T}) = \sum_{Q \in \mathcal{T}} \deg Q.$$

If $L_a > D(\mathcal{T})$, then G_a possesses an irreducible factor outside $\mathcal{T}$.

Proof. If every factor belonged to $\mathcal{T}$, then $A_a \leq D(\mathcal{T})$, contradicting Theorem 3.1. □

Corollary 5.3 (Asymptotic squarefreeness at fixed prime support)**.** *Let (u_n, v_n) be coprime pairs whose ordinary prime supports lie in a fixed finite set, and suppose*

$$M_n := \max\{h(u_n), h(v_n)\} \longrightarrow \infty.$$

For either sign, provided $f_{u_n} \pm f_{v_n} \neq 0$, set $G_n = f_{u_n} \pm f_{v_n}$. Then

$$\frac{\deg \operatorname{rad}(G_n)}{\deg G_n} \longrightarrow 1.$$

Proof. The relevant two-term Mason argument gives

$$\mu(G_n) \leq \rho(u_n) + \rho(v_n) - 1 = O(1),$$

because the prime support is fixed. By the cancellation estimate, $\deg G_n \geq M_n - O(1) \to \infty$. Hence $\deg \operatorname{rad}(G_n)/\deg G_n = 1 - \mu(G_n)/\deg G_n \to 1$. □

6 Exponent bounds and finite-support finiteness

Corollary 6.1 (Explicit ordinary prime-exponent bounds)**.** *If $p^e \mid a$, then*

$$e \leq \frac{\min\{\rho(a) + \rho(b) + A_c - 1,\ \rho(a) + \rho(c) + A_b - 1\}}{m_2(p)}.$$

The corresponding cyclic estimates hold for prime powers dividing b or c.

Proof. Complete additivity gives $e\,m_2(p) \leq h(a)$. The two inequalities of Theorem 2.1 containing $h(a)$ give the displayed upper bound for $h(a)$. □

Remark 6.2 (One-sided control)**.** To bound all exponents in a, it is enough to control A_b and A_c; no information on A_a is required. This can be useful when only two of the three lift polynomials are computationally accessible.

Theorem 6.3 (Explicit S-unit finiteness)**.** *Fix a finite set of primes S and a real number $B \geq 0$. Put*

$$R_S := \sum_{p \in S} m_2(p).$$

There are only finitely many primitive triples $a + b = c$ with $ab > 1$, all prime factors of abc in S, and

$$A_a, A_b, A_c \leq B.$$

More precisely, for each $p \in S$,

$$v_p(a), v_p(b), v_p(c) \leq \left\lfloor \frac{R_S + B - 1}{m_2(p)} \right\rfloor.$$

Before imposing $a + b = c$, the number of possible assignments of prime powers to the three pairwise coprime entries is at most

$$\prod_{p \in S} \left(1 + 3 \left\lfloor \frac{R_S + B - 1}{m_2(p)} \right\rfloor \right).$$

Proof. For example,

$$h(a) \leq \rho(a) + \rho(b) + A_c - 1 \leq R_S + B - 1.$$

The same holds for b, c. Now apply $v_p(n) m_2(p) \leq h(n)$. Pairwise coprimality means each prime is assigned to at most one of a, b, c, giving the stated crude count. $\square$

Corollary 6.4 (Double S-unit finiteness)**.** *Fix, in addition, finite sets of irreducibles $\mathcal{T}_a, \mathcal{T}_b, \mathcal{T}_c$. There are only finitely many primitive triples with prime support in S and*

$$\operatorname{supp} \operatorname{rad}(G_i) \subseteq \mathcal{T}_i \qquad (i = a, b, c).$$

Proof. The support condition gives $A_i \leq \sum_{Q \in \mathcal{T}_i} \deg Q$. Apply Theorem 6.3. $\square$

Corollary 6.5 (Prime-power triples)**.** *Suppose*

$$a = p^r, \qquad b = q^s, \qquad c = \ell^t$$

for distinct primes p, q, ℓ. Writing $w_p = m_2(p)$ and similarly for q, ℓ, one has

$$A_c \geq \max\{r w_p, s w_q\} - w_p - w_q + 1,$$

and, after ordering $r w_p, s w_q, t w_\ell$ as $L_{(1)} \leq L_{(2)} \leq L_{(3)}$,

$$L_{(2)} + 2L_{(3)} \leq 2(w_p + w_q + w_\ell) + A_\Sigma - 3.$$

7 Fermat–Catalan, Catalan, and Pillai forms

Theorem 7.1 (Modified Pratt–Fermat–Catalan inequality)**.** *Let*

$$x^r + y^s = z^t$$

be a primitive solution with $x, y, z > 1$ and $r, s, t \geq 1$. Define

$$L := \max\{r h(x), s h(y), t h(z)\},$$

and let L_2 be the second largest of these three weighted heights. Let A_Σ be the sum of the radical degrees of

$$f_z^t - f_y^s, \qquad f_z^t - f_x^r, \qquad f_x^r + f_y^s.$$

Then

$$\boxed{2L \left(1 - \frac{1}{r} - \frac{1}{s} - \frac{1}{t} \right) + L_2 \leq A_\Sigma - 3.}$$

Proof. Apply (S3R) to $a = x^r$, $b = y^s$, $c = z^t$. Since $f_{x^r} = f_x^r$,

$$h(x^r) = rh(x), \qquad \rho(x^r) = \rho(x),$$

and similarly for y, z. Thus

$$L_2 + 2L \leq 2\big(\rho(x) + \rho(y) + \rho(z)\big) + A_\Sigma - 3.$$

Because $\rho(n) \leq h(n)$,

$$\rho(x) + \rho(y) + \rho(z) \leq \frac{L}{r} + \frac{L}{s} + \frac{L}{t}.$$

Rearrange. □

Corollary 7.2 (Consequences at and below the Fermat–Catalan threshold)**.** *In the setting of Theorem 7.1:*

(a) If $1/r + 1/s + 1/t < 1$ and $A_\Sigma \leq B$, then L is explicitly bounded.

(b) If $1/r + 1/s + 1/t = 1$, then $L_2 \leq A_\Sigma - 3$.

(c) Along any family with $L \to \infty$ and $A_\Sigma = o(L)$,

$$\liminf \left(\frac{1}{r} + \frac{1}{s} + \frac{1}{t}\right) \geq 1.$$

Corollary 7.3 (Catalan-type form)**.** *For a primitive equation*

$$1 + y^s = x^r$$

with $x, y > 1$, let A_Σ be the three-radical sum associated with $(1, y^s, x^r)$. Then

$$2\max\{rh(x), sh(y)\} + \min\{rh(x), sh(y)\} \leq 2\big(\rho(x) + \rho(y)\big) + A_\Sigma - 3.$$

Thus bounded A_Σ forces explicit bounds on the weighted exponents.

Corollary 7.4 (Pillai-type form)**.** *For a primitive equation*

$$k + y^s = x^r$$

with fixed k, the same argument gives an explicit bound involving $\rho(k)$, $h(k)$, and the three additive radical degrees. Consequently, for fixed prime support and bounded three-radical complexity, only finitely many exponent pairs (r, s) can occur.

8 Power sums, power differences, and fixed factor sets

Theorem 8.1 (The families $f_n^k \pm 1$)**.** *Let $n > 1$ and $k \geq 1$. Then*

$$\deg \operatorname{rad}(f_n^k \pm 1) \geq kh(n) - \rho(n) + 1,$$
$$\deg \gcd(f_n^k \pm 1, (f_n^k \pm 1)') \leq \rho(n) - 1.$$

Proof. For the plus sign, apply Mason–Stothers to the triple $(f_n^k, 1, f_n^k + 1)$. For the minus sign, apply it to the triple $(f_n^k - 1, 1, f_n^k)$. In either case the nonconstant term has degree $kh(n)$ and radical degree $\rho(n)$, while the constant term contributes nothing. Rearrangement gives the first inequality; subtracting the radical degree from the full-degree Mason estimate gives the second. □

Corollary 8.2 (Infinitely many factors and uniform multiplicity bounds)**.** *For every fixed $n > 1$,*

$$\bigcup_{k\geq 1} \mathrm{Irr}(f_n^k - 1) \quad \textit{and} \quad \bigcup_{k\geq 1} \mathrm{Irr}(f_n^k + 1)$$

are infinite. Moreover, for every fixed irreducible Q and every k,

$$v_Q(f_n^k \pm 1) \leq 1 + \left\lfloor \frac{\rho(n) - 1}{\deg Q} \right\rfloor.$$

Proof. The radical degree tends to infinity with k, so a finite union of possible factors is impossible. The multiplicity estimate follows from the second inequality in Theorem 8.1. ☐

Theorem 8.3 (General power sums and differences)**.** *Let $u, v > 1$ be coprime and $r, s \geq 1$. For either sign, provided the polynomial is nonzero,*

$$\deg \mathrm{rad}(f_u^r \pm f_v^s) \geq \max\{rh(u), sh(v)\} - \rho(u) - \rho(v) + 1,$$
$$\deg \gcd(f_u^r \pm f_v^s, (f_u^r \pm f_v^s)') \leq \rho(u) + \rho(v) - 1.$$

Proof. For the plus sign, apply Mason to $f_u^r + f_v^s$. For the minus sign, use the relation $(f_u^r - f_v^s) + f_v^s = f_u^r$ (or its symmetric version). The two base polynomials are coprime, their degrees are $rh(u), sh(v)$, and their radical degrees are $\rho(u), \rho(v)$. ☐

Corollary 8.4 (Finiteness with prescribed polynomial factors)**.** *Fix coprime $u, v > 1$ and a finite set $\mathcal{T}$ of irreducibles. There are only finitely many exponent pairs (r, s) for which all irreducible factors of $f_u^r \pm f_v^s$ lie in $\mathcal{T}$.*

Proof. If all factors lie in $\mathcal{T}$, then the radical degree is at most $D(\mathcal{T})$. Apply Theorem 8.3 to bound $rh(u)$ and $sh(v)$. ☐

9 All Pratt coordinates

For a prime q, write $d_q := m_2(q)$. The polynomial Pratt coordinate indexed by f_q recovers the numerical coordinate $m_q(n)$, and the terminal-capacity inequality gives $d_q m_q(n) \leq h(n)$ [1, Sections 15–18].

Theorem 9.1 (Coordinatewise symmetric three-radical inequality)**.** *Order the three numbers $m_q(a), m_q(b), m_q(c)$ as $u_{(1)} \leq u_{(2)} \leq u_{(3)}$. Then*

$$\boxed{d_q\big(u_{(2)} + 2u_{(3)}\big) \leq 2\rho(abc) + A_\Sigma - 3.}$$

Proof. The coordinate form of the cyclic Mason inequalities gives

$$d_q \max\{m_q(b), m_q(c)\} \leq \rho(b) + \rho(c) + A_a - 1,$$
$$d_q \max\{m_q(a), m_q(c)\} \leq \rho(a) + \rho(c) + A_b - 1,$$
$$d_q \max\{m_q(a), m_q(b)\} \leq \rho(a) + \rho(b) + A_c - 1.$$

Add and use the same pairwise-maximum identity as in Theorem 2.2. ☐

Corollary 9.2 (Ordinary q-adic exponents)**.** *Since $v_q(n) \leq m_q(n)$,*

$$d_q v_q(a) \leq \min\{\rho(a) + \rho(b) + A_c - 1,\ \rho(a) + \rho(c) + A_b - 1\},$$

with cyclic analogues for b and c.

Corollary 9.3 (Compactness of recursive ancestry profiles)**.** *For fixed ordinary prime support and bounded A_a, A_b, A_c, only finitely many complete Pratt-coordinate vectors $M(a), M(b), M(c)$ can occur. Equivalently, only finitely many labelled Pratt forests occur.*

10 Complete-front probability consequences

The complete-front interpretation identifies $f_n(x)/f_n(1)$ as a probability generating function, and the mean, variance, and every cumulant of the terminal-leaf statistic are completely additive arithmetic functions [2]. Write $\kappa_j(n)$ for any fixed cumulant.

Corollary 10.1 (Bounded complete-front statistics)**.** *Fix a finite prime set S and a bound B for A_a, A_b, A_c. Among primitive triples with prime support in S, all quantities*

$$\mathbb{E}X_n, \qquad \operatorname{Var}(X_n), \qquad \kappa_j(n)$$

for $n \in \{a, b, c\}$ and fixed j are uniformly bounded.

Proof. By Theorem 6.3, all exponents $v_p(n)$ are bounded. Complete additivity gives $\kappa_j(n) = \sum_{p\in S} v_p(n)\kappa_j(p)$, and similarly for the mean and variance. □

Remark 10.2 (Combinatorial meaning of radical growth)**.** The height $h(n)$ is the number of terminal 2-vertices in the full Pratt forest, while $\rho(n)$ counts the terminal weights contributed by distinct root-prime types. Thus Theorem 3.1 says that repeated terminal complexity in two input forests must be compensated by distinct irreducible factors in their additive polynomial lift.

11 Hadamard–Pratt powers

The Hadamard–Pratt product is defined by

$$m_p(x \star_P y) = m_p(x)m_p(y),$$

and the natural Pratt cone is closed under this operation [3]. Define $n^{\star_P k}$ by iterating $\star_P$. Then

$$m_p(n^{\star_P k}) = m_p(n)^k, \qquad h(n^{\star_P k}) = h(n)^k.$$

The full Pratt support stays fixed. Put

$$C(n) := \sum_{p\in \operatorname{supp} M(n)} m_2(p).$$

Since the ordinary prime support is contained in the full Pratt support, $\rho(n^{\star_P k}) \le C(n)$.

Theorem 11.1 (Exponential radical growth for Hadamard–Pratt powers)**.** *Let $h(n) > 1$. Then*

$$\deg \operatorname{rad}\big(f_{n^{\star_P k}} \pm 1\big) \ge h(n)^k - C(n) + 1,$$
$$\deg \gcd\big(f_{n^{\star_P k}} \pm 1, (f_{n^{\star_P k}} \pm 1)'\big) \le C(n) - 1.$$

Consequently,

$$\frac{\deg \operatorname{rad}(f_{n^{\star_P k}} \pm 1)}{\deg f_{n^{\star_P k}}} \longrightarrow 1$$

with an exponentially decaying defect.

Proof. Apply Theorem 8.1 with exponent 1 to the integer $n^{\star_P k}$, then use $h(n^{\star_P k}) = h(n)^k$ and $\rho(n^{\star_P k}) \le C(n)$. □

12 Ramification and monodromy-oriented consequences

For a nonconstant polynomial G, the quantity

$$\mu(G) = \deg \gcd(G, G') = \sum_{G(\alpha)=0} (e_\alpha(G) - 1)$$

is the total excess multiplicity in the fibre over 0.

Corollary 12.1 (Total ramification excess in the three zero fibres)**.** *For all nonconstant members among G_a, G_b, G_c,*

$$\boxed{\sum_{i\in\{a,b,c\}} \sum_{G_i(\alpha)=0} (e_\alpha(G_i) - 1) \leq 2\rho(abc) - 3.}$$

Constant G_i contribute zero.

Proof. This is Theorem 4.1 interpreted over an algebraic closure. □

Remark 12.2 (Potential Galois use)**.** In fixed-prime-support families with growing heights, the degrees of the lift polynomials can grow while the total multiple-root defect remains bounded. Their zero fibres are therefore asymptotically dominated by simple points. This does not by itself prove full symmetric monodromy, but it supplies useful input for discriminant and branch-cycle tests of the kind developed for Pratt polynomials in [4].

13 Computational consequences and extremal diagnostics

Corollary 13.1 (Certificate for an incomplete factorisation)**.** *Suppose distinct irreducible factors of G_a have been found with total degree D_{known}. Then the as-yet-unfound squarefree part has degree at least*

$$\max\Big\{0, \max\{h(b), h(c)\} - \rho(b) - \rho(c) + 1 - D_{\text{known}}\Big\}.$$

Corollary 13.2 (Search-space pruning)**.** *If a search is restricted by $A_a, A_b, A_c \leq B$, every candidate must satisfy*

$$h_{(2)} + 2h_{(3)} \leq 2\rho(abc) + 3B - 3.$$

This test uses only integer Pratt data before any factorisation of the large lift polynomials.

Definition 13.3 (Three-radical quality and cyclic slacks)**.** Define

$$Q_3(a, b, c) := \frac{h_{(2)} + 2h_{(3)}}{2\rho(abc) + A_\Sigma - 3}$$

and

$$\sigma_a := \rho(b) + \rho(c) + A_a - 1 - \max\{h(b), h(c)\},$$
$$\sigma_b := \rho(a) + \rho(c) + A_b - 1 - \max\{h(a), h(c)\},$$
$$\sigma_c := \rho(a) + \rho(b) + A_c - 1 - \max\{h(a), h(b)\}.$$

Proposition 13.4 (Extremal diagnostics)**.** *For every admissible triple,*

$$0 < Q_3(a, b, c) \leq 1,$$

and the exact total slack identity is

$$\sigma_a + \sigma_b + \sigma_c = 2\rho(abc) + A_\Sigma - 3 - \big(h_{(2)} + 2h_{(3)}\big).$$

Thus equality in (S3R) holds exactly when all three cyclic Mason inequalities are sharp.

Proof. Nonnegativity of each σ_i is Theorem 2.1. Summing gives the identity. The quality bound is equivalent to (S3R). □

14 Summary of the application spectrum

The symmetric theorem packages three ordinary Mason–Stothers applications into one arithmetic statement. Its principal uses can be grouped as follows:

Theme	Consequence
Radical growth	Lower bounds for each A_i and for A_Σ; balanced triples force the largest total radical.
Cancellation	The order of cancellation at infinity is bounded solely by squarefree Pratt data.
Multiple roots	$D_a + D_b + D_c \leq 2\rho(abc) - 3$ and uniform multiplicity bounds for every irreducible factor.
Perfect powers	A k-th power lift forces the relevant heights below $\frac{k}{k-1}$ times the pairwise Pratt radical.
Factorisation	Either many factors occur or one factor has large degree; finite factor sets must eventually be escaped.
Asymptotics	Fixed-prime-support lift families become asymptotically squarefree.
Diophantine finiteness	Explicit exponent bounds and finite S-unit families under bounded three-radical complexity.
Fermat–Catalan	A defect-corrected reciprocal-exponent inequality with the classical threshold $1/r + 1/s + 1/t = 1$.
Power sequences	Linear radical growth for $f_n^k \pm 1$ and $f_u^r \pm f_v^s$, with bounded repeated-root mass.
All coordinates	Coordinatewise S3R bounds the complete recursive ancestry profile, not only the terminal coordinate.
Probability	Fixed support and bounded three-radical complexity imply bounded complete-front cumulants.
Hadamard arithmetic	Hadamard–Pratt powers produce exponential radical growth and exponentially fast asymptotic squarefreeness.
Geometry	The total ramification excess in the three zero fibres is bounded arithmetically.
Computation	Partial-factorisation certificates, search pruning, a quality ratio, and exact cyclic slacks.

15 Conclusion

The inequality (S3R) is elementary once the Pratt-polynomial dictionary is available: apply Mason–Stothers to the three cyclic polynomial lifts and add. Its usefulness comes from the fact that the left side measures repeated recursive arithmetic complexity, while the right side separates squarefree Pratt support from genuinely additive polynomial complexity. The resulting balance controls factorisation, multiplicity, exponent growth, finite-support families, power equations, recursive coordinates, probability distributions on complete fronts, Hadamard–Pratt dynamics, and ramification.

The most structurally important companion statement is the multiple-root defect estimate

$$D_a + D_b + D_c \leq 2\rho(abc) - 3,$$

which shows that additive Pratt lifts can have very large degree while retaining only a bounded amount of repeated-root mass. This makes the three-radical theorem a practical bridge from the Pratt-coordinate proof of Mason–Stothers to a broad family of new, provable arithmetic and algebraic consequences.

References

[1] Orges Leka, *Pratt Coordinates for Natural Numbers and Monic Rational Polynomials: Reconstruction, Hilbert-Space Embeddings, and a Pratt-Coordinate Form of Mason–Stothers*, Version 0.2, August 3, 2026. Supplied file: `pratt_polynomial_coordinates(3).pdf`.

[2] Orges Leka, *Complete Node Fronts of Pratt Forests: Generating Polynomials and a Probabilistic Interpretation of* $f_n(x)$, August 5, 2026. Supplied file: `Complete_Node_Fronts_of_Pratt_Forests_Probabilistic(1).pdf`.

[3] Orges Leka, *Hadamard–Pratt Arithmetic: A Coordinatewise Product on Pratt Coordinates, a Semiring on* $\mathbb{N}$*, and the Spectrum of Its Unitization*, Draft, August 4, 2026. Supplied file: `Hadamard-Pratt_Arithmetic(1).pdf`.

[4] Orges Leka, *Root-Label Fields on Pratt Forests: An Application-First Multivariate Theory of Complete Fronts, Coordinate Transport, Arithmetic Fibres, and Monodromy*, Version 0.3, August 6, 2026. Supplied file: `Root_Label_Fields_on_Pratt_Forests_v0.3_empirically_verified(1).pdf`.